

Step-Detection-Controlled Robot Dog Guide for Visually Impaired Pedestrians

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Abstract—The integration of assistive technologies into the daily lives of visually impaired individuals shows significant promise for enhancing mobility and independence. This paper introduces a novel assistive device: a step-detection-controlled robot dog guide for visually impaired pedestrians. Utilizing a smartphone's accelerometer, the device detects the user's walking steps and synchronizes its movement with the user's pace through real-time data transmission, thereby ensuring an adaptive and responsive navigation aid. To achieve reliable step detection, we employ a threshold-based algorithm enhanced by peak detection techniques. This approach enables the robot to accurately match the walking speed and pattern of the user. The smartphone and the robot are able to communicate seamlessly using Bluetooth technology. This enables the user to control the robot's start, stop, and maneuvering functions in response to their movements and commands. Extensive preliminary tests conducted in a laboratory environment demonstrate the robot's capability to effectively detect and adapt to changes in the user's pace while navigating typical urban obstacles. Furthermore, the findings suggest that the step-detection-controlled robot dog significantly aids visually impaired pedestrians in navigating complex environments. This research opens promising avenues for future developments in robotic assistance, underscoring the potential for broader applications and enhancements.

Index Terms—Robot Dog, Visually Impaired, Smartphone, Indoor Navigation, Android

I. INTRODUCTION

Assistive technologies designed for navigating complex environments have the potential to significantly improve the mobility and independence of visually impaired individuals. Traditional aids such as canes and guide dogs provide valuable service but come with limitations in terms of information processing and environmental adaptability [1]. Conversely, the recent advancements in robotics and sensor technology present an opportunity to augment these traditional aids with more adaptive and responsive solutions [2]. On the other hand, according to the World Health Organization, as of 2023, there are at least 2.2 billion people globally with near or distance vision impairments [3]. Of these, at least 1 billion cases could have been prevented or are yet to be addressed. This marks a substantial increase from the 2012 estimate, which recorded 314 million individuals with vision impairments.

These statistics underscore the growing need for more adaptive and responsive solutions in assistive technologies.

Guide dog is well-known as assistance animal for the visually impaired. It is able to alert its owner to special emergencies and help him avoid obstacles. However, they need extensive training over a long period of time, which can cost and exceeds to \$50,000 [4], in order to adapt to their owners' daily routines and lifestyles. Additionally, they can only provide guidance for 8 to 10 years. Therefore, finding a solution that allows the visually impaired to navigate safely and affordably is worthwhile.

On the other hand, the development of assistive technologies for visually impaired individuals has been a critical area of research within the field of robotics and artificial intelligence. A pioneering concept in this domain is the robotic guide dog, designed to offer navigation assistance in various environments. This initiative addresses the need to provide alternatives to traditional guide dogs, which, while effective, come with challenges related to training, cost, and availability.

Indeed, the significant progress in sensor technology, machine learning, and mobile robotics has led to the development of these robotic systems [5]. Initial research in this area concentrated on simple obstacle avoidance methods utilizing ultrasonic and infrared sensors. For example, a study conducted by the authors of [6] investigated the application of ultrasonic sensors for identifying obstacles at various heights and distances, demonstrating their importance for indoor navigation purposes.

Continuing on this path, recent studies have integrated advanced computer vision techniques to interpret intricate environments. These systems use cameras and deep learning algorithms not only to detect obstacles but also to identify them, enabling the robot to provide contextual information to the user. For instance, Bruno et al [7], conducted research that integrated real-time object recognition to enhance the environmental understanding of the robotic guide. Additionally, the integration of tactile and auditory feedback mechanisms has also been a focus to improve the user experience [8]. These enhancements help in translating the visual information

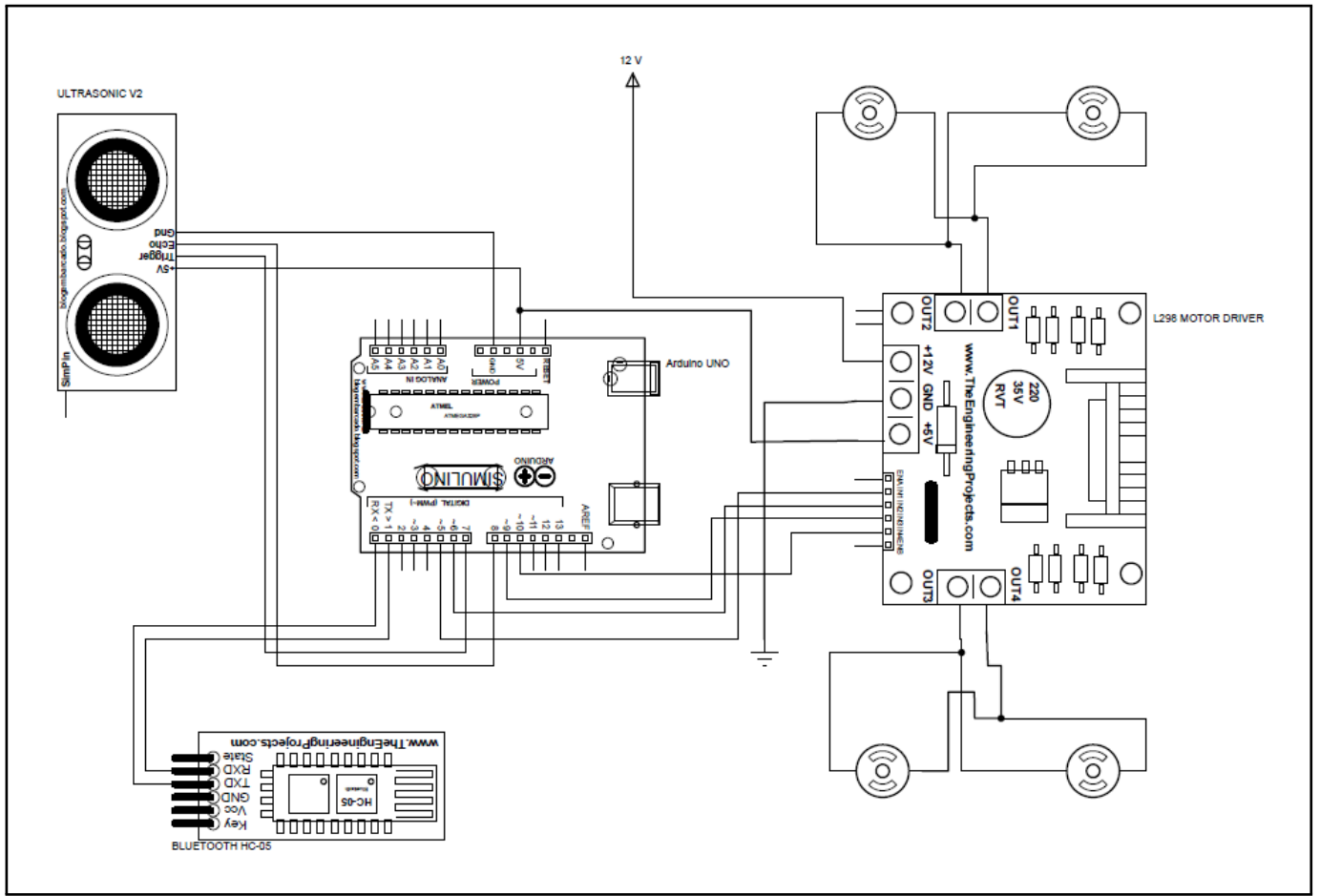


Fig. 1. Configuration of the proposed Robot Dog Guide System.

captured by the robot into understandable formats for visually impaired users, thus facilitating smoother and more intuitive navigation.

This paper presents a novel assistive device designed to address these challenges: a robot dog guide controlled by a step-detection approach. This robotic guide utilizes the accelerometer of a smartphone to detect the walking steps of a user, synchronizing its movement to match the pace of the user in real-time. Such synchronization promises to provide a more intuitive and seamless navigation experience for visually impaired pedestrians. This innovative approach exemplifies the potential of combining robotics and sensor technology to enhance traditional aids for individuals with visual impairments.

The rest of the paper is organized as follows. In Section II, the functional principles of the proposed robot guide system are described. Section III presents an Android application that remotely interacts in real-time with the robot dog system. In Section IV, the results of experiments carried out in this work are presented and discussed. Finally, Section V concludes the paper, summarizing the key findings and discussing their implications for the field of assistive robotics.

II. CONFIGURATION OF THE PROPOSED ROBOT DOG GUIDE SYSTEM

The Robot Dog Guide System is specifically designed to improve indoor navigation for individuals with visual impairments, with a focus on improving mobility and safety. This system includes a mobile robot platform powered by four DC motors, which are driven by a 12V power supply through an L298 Motor Driver Module. The key components of the system include an Arduino Uno board equipped with a Bluetooth module for wireless communication and an Ultrasonic Sensor. The DC motors and the Arduino board are powered by a 12V battery and a 9V battery, respectively. These components are illustrated within Fig. 1.

For remote step detection, the system utilizes accelerometer data from a smartphone. Due to the inherent complexity and noise associated with accelerometer data, it is processed using an Exponential Moving Average (EMA) filter on the Android Platform. The EMA filter is a recursive filter that effectively smooths noisy data. The filter equation is given by the Eq. (1) [9].

$$y(t) = \alpha.x(t) + (1 - \alpha).y(t - 1) \quad (1)$$

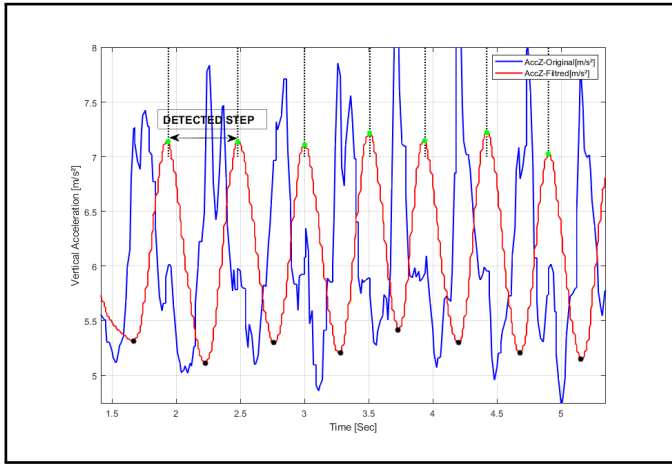


Fig. 2. Step detection principle.

Where $y(t)$ represents the output at time t , $x(t)$ denotes the input at time t , α is the smoothing factor (a constant between 0 and 1), and $y(t - 1)$ is the output from the previous time step.

The filtered data are subsequently analyzed to detect steps using peak detection methods [10], as shown in Fig. 2. Once a step is detected, a simple command drives the robot forward. If the robot's ultrasonic sensor detects an obstacle, it immediately stops and sends a notification to the smartphone app alerting the user of the presence of the obstacle. Section III will describe the Android application developed specifically for this feature.

III. ROBOT DOG GUIDE: AN ANDROID-BASED APPLICATION

Recently, various assistive technologies have been created to help impaired individuals become more independent using mobile applications. For instance, in [11] a system that combines sensors and an Android application to assist users in navigating around obstacles is proposed. In this work, a new application has been developed to control a robot guide system as detailed in Section II. As shown in Fig. 3, the application uses Bluetooth to connect to the robot allowing users to turn it on or off and pair it with their smartphone. In addition to managing connectivity, the application can monitor the status of the robot and guide its movements while tracking the user's steps to synchronize with their pace. Furthermore, the application calculates the distance traveled providing real-time updates on the journey progress.

Utilizing an Ultrasonic sensor embedded within the robot, our Android-based application empowers your robot dog to intelligently detect obstacles and navigate around them with precision. Instant voice alerts ensure swift notification to users, guaranteeing safe navigation in diverse environments. Equipped to confidently maneuver through different surroundings, your robot dog provides peace of mind and seamless interaction. Fig. 4 illustrates the process flow of our application, detailing how it controls various functions to enhance

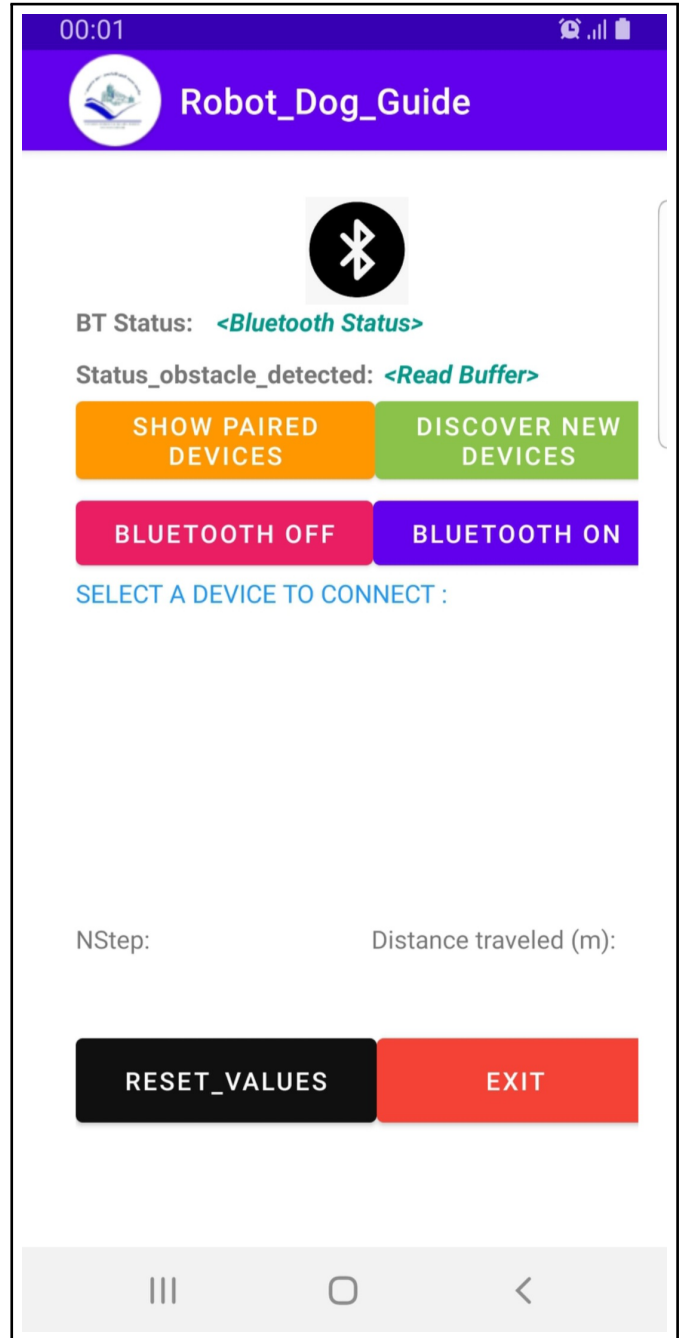


Fig. 3. A screenshot of Step-Tracking Robot Guide Application.

the user experience. From initiating commands to the robot dog, to receiving real-time feedback and adjusting settings, every interaction is seamlessly orchestrated to ensure intuitive control and maximum enjoyment for the user.

IV. EXPERIMENTAL RESULTS

The effectiveness of the Robot Dog Guide System was evaluated through a series of experiments designed to test its responsiveness and navigational assistance capabilities for visually impaired pedestrians, specifically in an indoor envi-

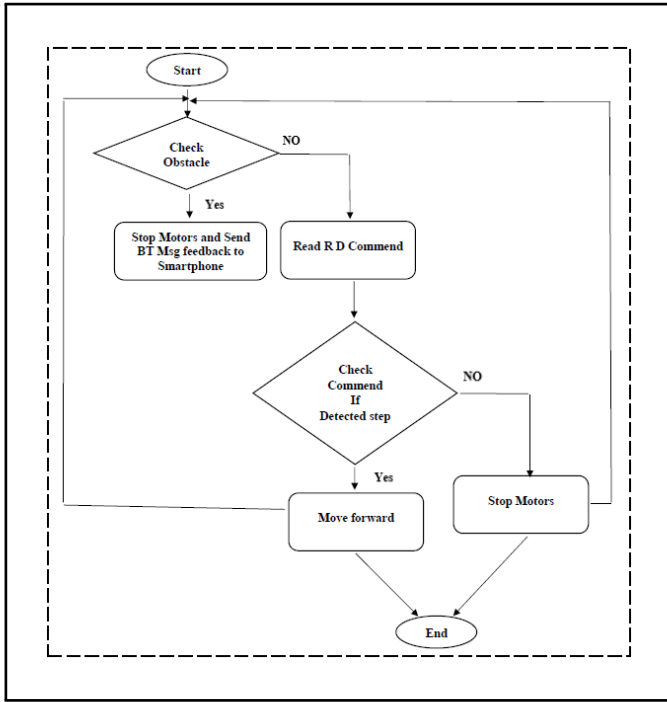


Fig. 4. Flowchart Robot Dog Guide Functioning.

ronment. These tests aimed to confirm the reliability of step detection and the system's ability to synchronize with the user's pace, as well as its obstacle detection and avoidance functionalities. Fig. 5 represents a real-world illustration of the realized robot dog guide system. The testing utilized a mobile robot platform equipped with an ultrasonic sensor for obstacle detection, while wireless communication between the Arduino Uno board and the smartphone was facilitated by an HC-05 Bluetooth module. The focus of the testing was on validating the developed Android application, which processed step detection through a threshold-based algorithm enhanced with peak detection, ensuring precise movement synchronization with the user's varying walking speeds.

Indeed, the results demonstrated that the robot could consistently match the user's pace, thereby validating the effectiveness of real-time data transmission for adaptive navigation. Concurrently, the robot was equipped with an ultrasonic sensor to test its obstacle-detection capabilities. During these tests, the robot successfully identified multiple obstacles and promptly halted movement to prevent potential collisions. This capability was crucial for ensuring the safety and enhancing the confidence of visually impaired users while navigating through typical indoor challenges, such as furniture, doorways, and stationary objects. These features demonstrate the system's ability to respond to user movement and environmental factors, making it an effective navigational aid for visually impaired pedestrians in indoor environments. The testing scenarios, as illustrated in Fig. 6 and 7, demonstrated the robot's responses to step synchronization and obstacle navigation, highlighting the integrated functionality of the system in dynamic real-

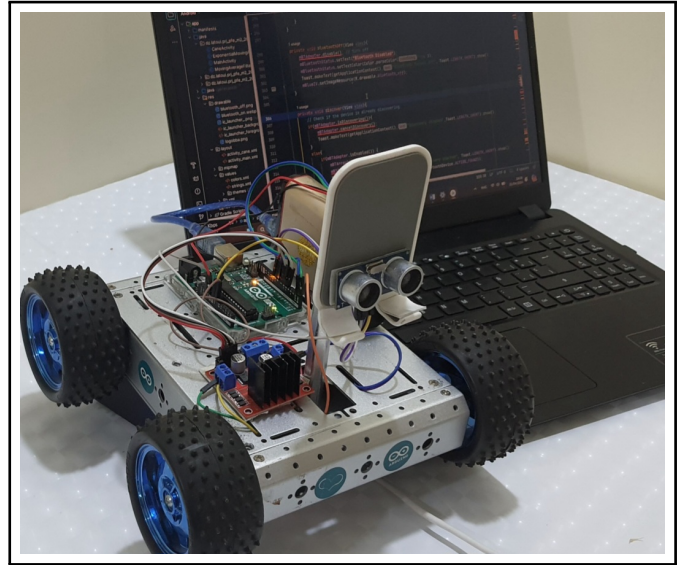


Fig. 5. Prototype guide dog for the visually impaired.

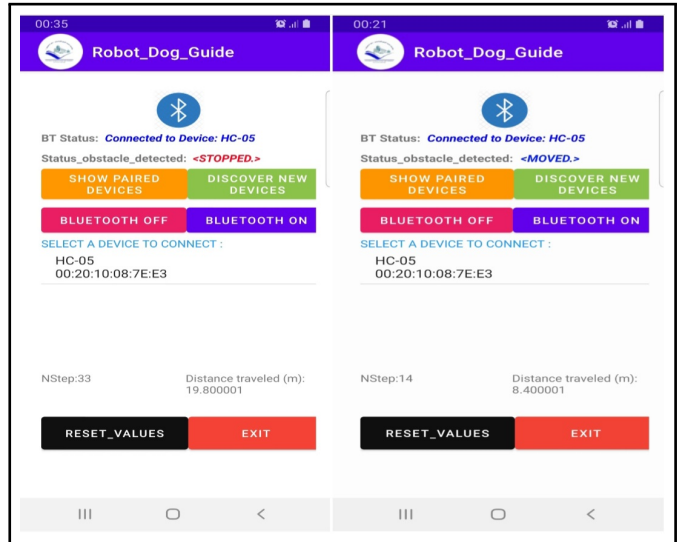


Fig. 6. Robot Guide App Interface Obstacle Detection Test Scenarios.

world conditions.

V. CONCLUSION

In this paper, a comprehensive evaluation of the proposed Robot Dog Guide System has successfully demonstrated its capability to enhance the mobility and safety of visually impaired individuals within indoor environments. Through rigorous testing, the system proved its proficiency in step detection, pace synchronization, and obstacle avoidance, which are critical for assisting visually impaired pedestrians. Integrating ultrasonic sensors and advanced processing algorithms enabled the system to navigate complex indoor environments effectively, adapting to various user movements and environmental challenges. The results of the experiments confirm that the system not only meets, but exceeds the requirements for a



Fig. 7. Testing Step-Controlled Robot Guide for the Visually Impaired.

reliable assistive device, offering a significant improvement over traditional aids like canes and guide dogs. The real-time data transmission for adaptive navigation has shown great potential in providing a more intuitive and seamless user experience, thereby fostering greater user independence. Moving forward, the insights gained from this project will serve as a valuable foundation for further development and refinement of robotic assistive technologies. Future research will focus on expanding the capabilities of the Robot Dog Guide System to outdoor environments, enhancing its obstacle detection algorithms, and integrating more user feedback to improve its interactive features. Ultimately, this research underscores the transformative potential of integrating advanced robotics with sensor technology to improve the quality of life for visually impaired individuals, paving the way for broader applications and societal benefits.

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